

Prototyping the next-generation spectrograph for space exploration: Digital Micromirror Device based Multi-object Spectrograph (DMD-MOS) calibration and performance assessment

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Digital Micromirror Device based Multi-Object Spectrograph (DMD-MOS) used a DMD to instantly configure multi-slits masks for the incoming telescope field, thus minimizing telescope overhead and facilitating efficient observing for hundreds of targets, which is an absolute game changer of next generation multi-objects spectrograph.

Introduction

- DMD:** A high precision micro-opto-electro-mechanical system with grids of actuated micromirrors. Allows for digital control of the orientation of each micromirror between **two opposite angles** and create multi-object slits.
- Incoming light is **split into two optical paths**:
 - Imaging channel:** Target acquisition & slit-mask creation
 - Spectral channel:** 400-700nm VPH spectrograph with $R \sim 1000$

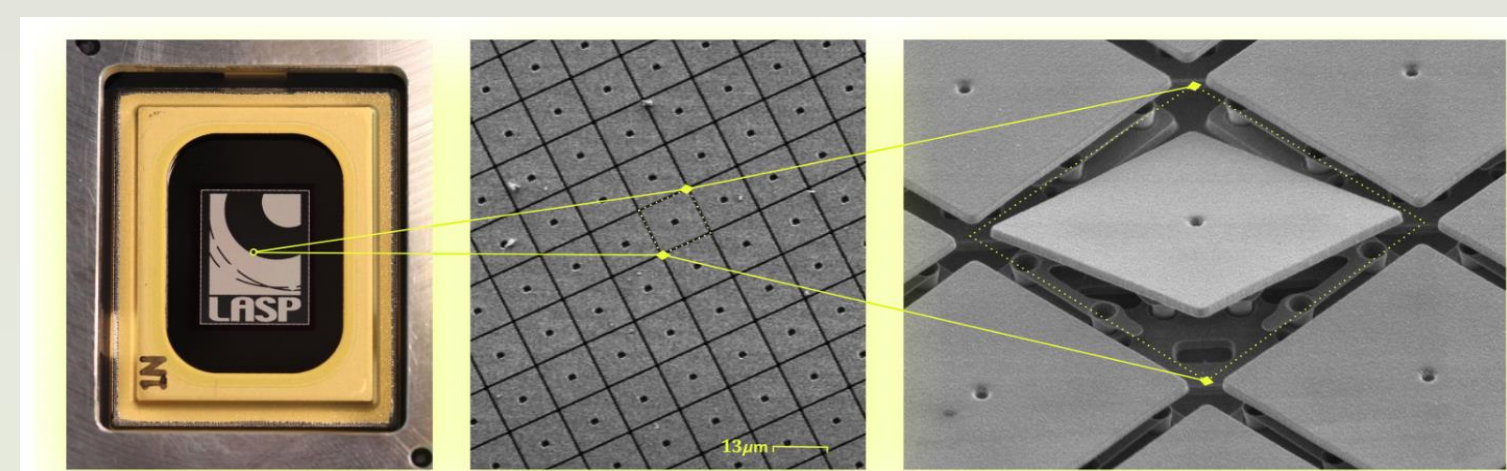


Fig 1: Zoomed in DMD with actuated micromirrors

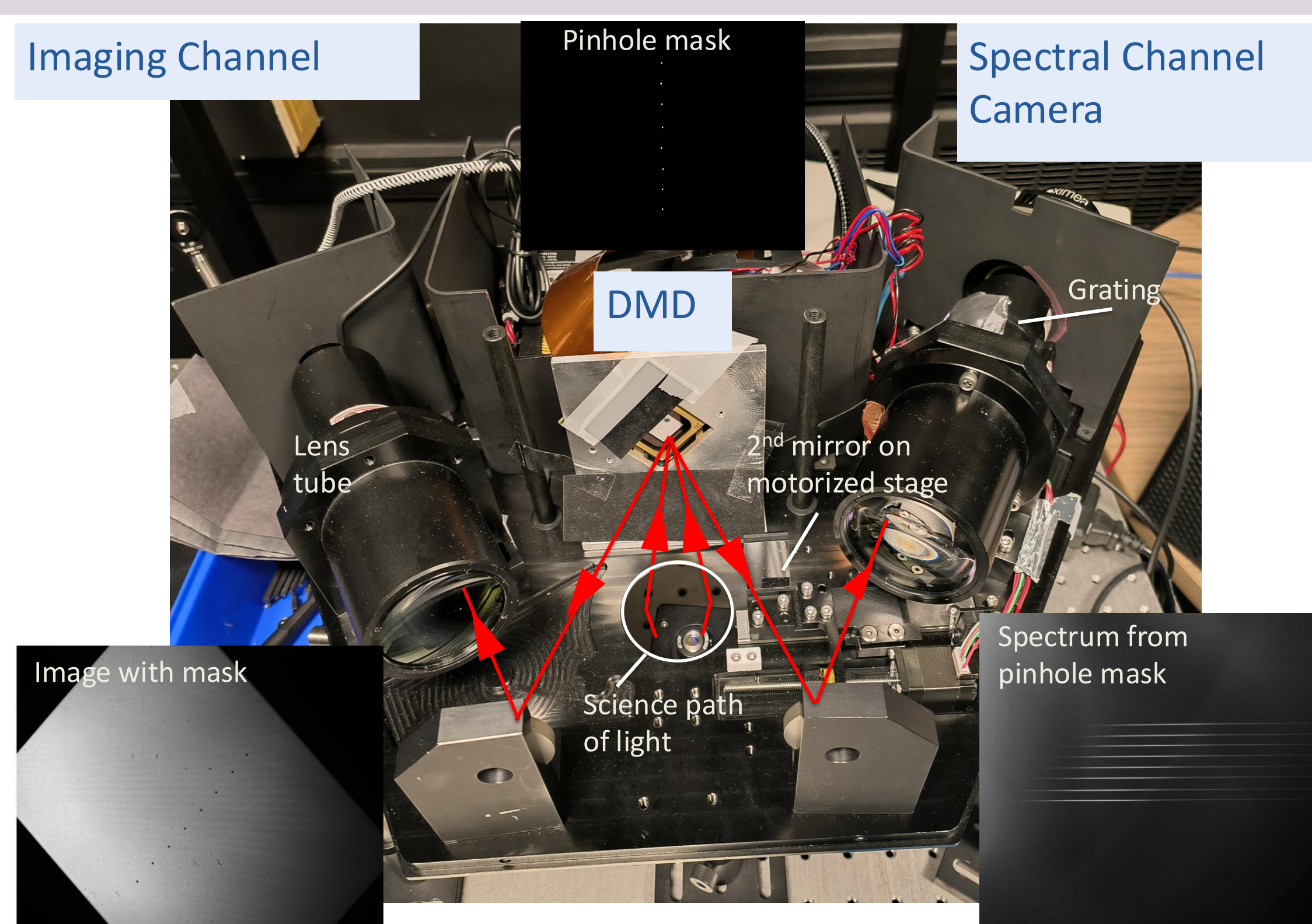


Fig 2: Set up of DMD-MOS and mask-imaging-spectrum relation.

- 99.62% accuracy^[1] with 50μs (20 kHz) DMD mirror flipping rate^[2]** leading >10 years lifetime^[1]. Show promise for future space missions.
- DMD array contains 768x1024 micromirrors with dimension **13x13μm** each, covers **10.5x13.4 arcmin FOV**. The **average DMD efficiency** across target wavelength range is **75%**.
- Motorized stage controls the calibration system, allow real-time flat-image capture and wavelength calibration.

Instrument performance

- Imaging quality of the emission lines was evaluated using different light sources [Ne, Hg, Ar, Kr & continuum light].
- The resulting emission lines from staircase slits have **PSF** along spectral direction **following a Gaussian curve**.
- For slit width of **18 micron** covering **1.15"** of the sky on DMD, the average FWHM such sky range is **32 micron** on spectral channel detector, **PSF and R meet design requirement**.
- Average wavelength dispersion is **0.25nm/px**.

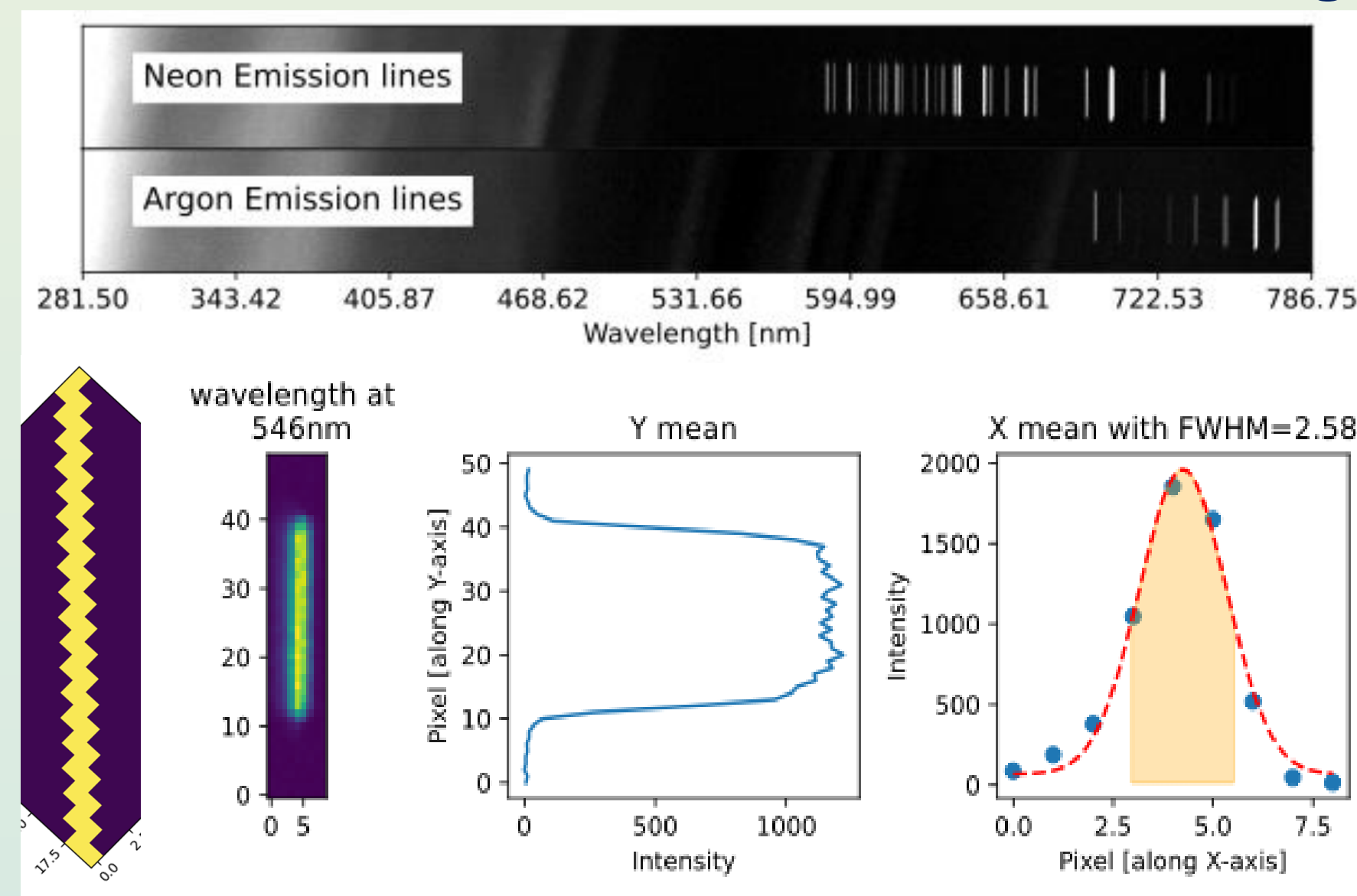


Fig 3: Emission lines of Neon and Argon arc lamp with wavelength calibrated [top]. DMD slit mask [bottom left]. Gaussian fit of the mean along x-direction of Neon emission line at 696.5nm [bottom right]

Mapping relation

- The precise DMD-imaging detector mapping is the key to allow us do accurate target selection in an instance from the entire FOV.
- Homography matrix** was used to match DMD pixels and Imaging channel pixels.

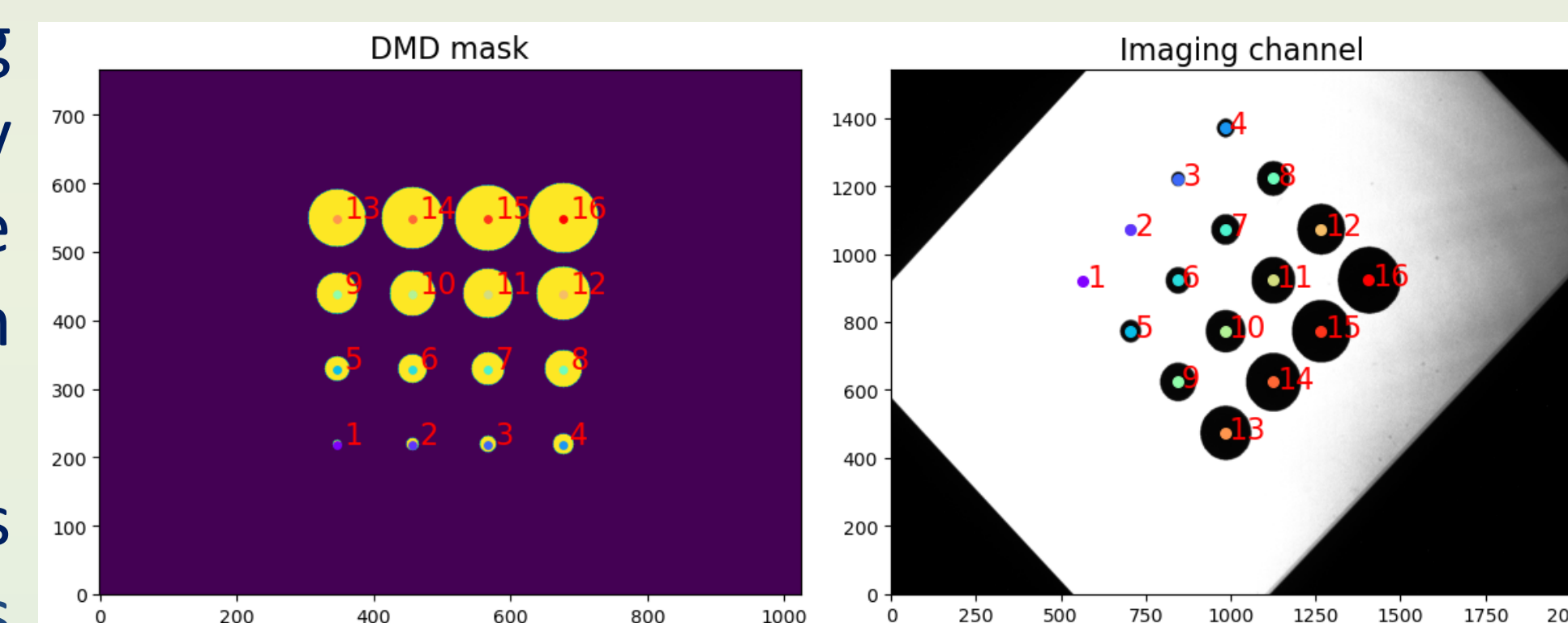


Fig 4: Centroid mapping between DMD-imaging channel by order of area.

- We matched the 16 centroids on DMD with 16 centroids on Imaging channel follow the order of area as the numbers labeled [Fig 4], mapping error is about 0.75 mirrors.
- Residuals were further evaluated [Fig 5], the position difference along y-direction is 0.3 in average with a **0.51 standard deviation**, along x-direction is 0.15 in average with **0.76 standard deviation**.

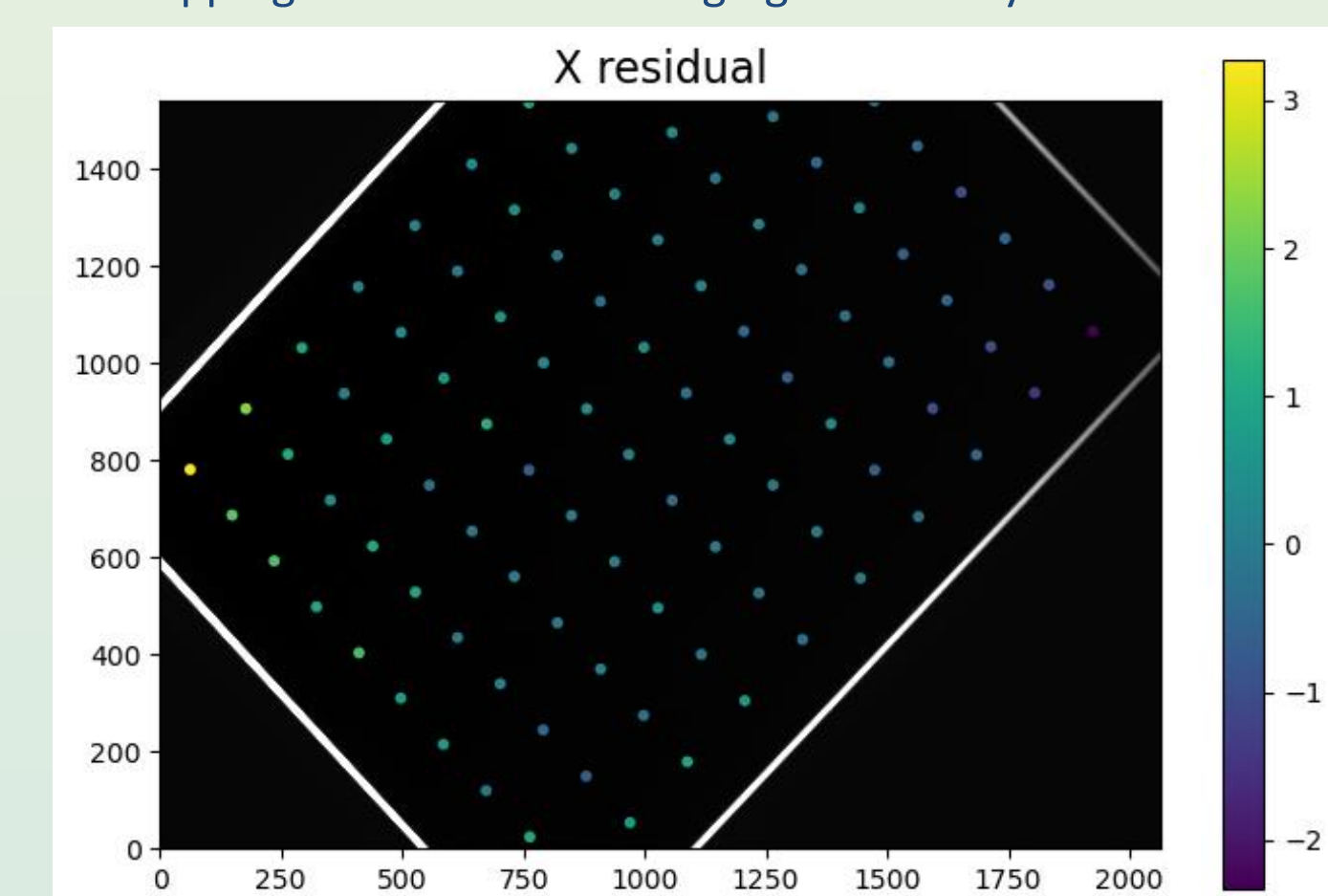


Fig 5: Position difference between the transferred DMD pinhole slits with homography matrix and the actual centroid of the pinhole signal from the imaging channel.

Sky test



Fig 6: DMD-MOS attached to CDK20-0.5m telescope.

- Sky test was performed in early June 2026 in Carr Astronomical Observatory (CAO), ON, Canada. DMD-MOS system was attached to a **CDK20-0.5m telescope** [Fig 6].
- Bright targets such as **Vega, Moon** were captured and processed during first observation with self-developed software & pipeline.
- Absorption lines of the Vega spectrum captured by DMD **matches** with reference spectrum using calibrated wavelength from lab after normalization [Fig 7], **approved the accuracy of wavelength calibration** of the system.
- Atmosphere transmission > 40%** (from ESO SkyCalc simulator), **total throughput is 20%-60%** for the system.

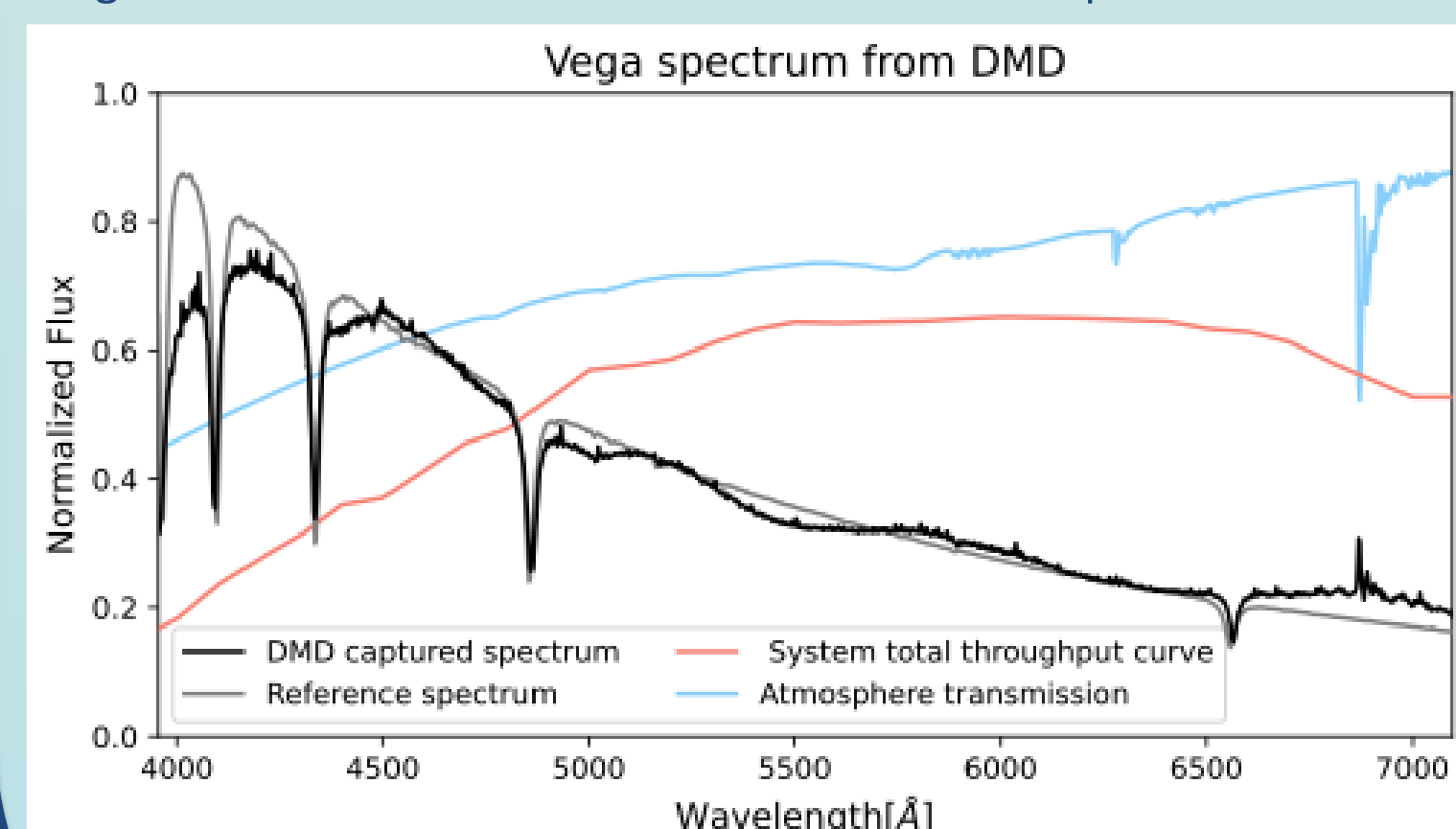


Fig 7: Comparison between reference Vega spectrum from Atlas³ (gray) With DMD captured spectrum (black). System throughput and ATM transmission^[4] plotted in light red and blue.

Summary & Future work

- Our DMD-MOS have a resolution of 1000 at center wavelength 550nm covers the wavelength range from 400-700nm, with flexible slits controls allow spectrum acquisition for multiple targets under 10.5'x13.4' FOV.
- Next step is to finishing up the on-sky commissioning tests including different type of stars and stars in open clusters.
- We also aim to exploring feasibility for future space missions.

Acknowledgement

- This work was partially supported by the Dunlap Seed Funding program at the University of Toronto. The author Jiayi (Emma) Xu would like to thank the DMD-MOS team for their collaborative effort and invaluable help.
- DMD_MOS Full Team: PI: Dr. Sarik Jeram, Dr. Shaojie Chen; Consultant: Prof. Ting Li, Prof. Suresh Sivanandam; Software: Zain Azam, Sarah He; System design: Mark Barnett, Daksh Singh; Pipeline testing: Yixuan (Chris) Cheng; Lab testing: Ziang (Frank) Chen, Adi Khandelwal, Dhruv Pandya; Telescope managing: Prof. Michael Reid as well as the Department of Astronomy and Astrophysics and the Dunlap Institute.

Reference

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- ESO SkyCalc Atmosphere transmission simulation. <https://www.eso.org/observing/etc/bin/gen/form?INS.MODE=swspectr+INS.NAME=SKYCALC>

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